

Lab report 9

1. Objective

The objective of this lab is to understand the concept of functions in C++, including function declaration, function prototype, function calling, and passing parameters to functions.

2. Apparatus / Software

- Computer System
 - C++ Compiler (Code::Blocks / Dev C++ / VS Code)
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3. Theory

A function is a named block of code that performs a specific task. It can take input values (parameters), process them, and return a result.

A function prototype is a declaration that informs the compiler about:

- Function name
- Return type
- Parameters

Syntax:

```
returnType functionName(parameters);
```

4. Lab Tasks Implementation

Task 1: Simple Function Without Parameters

```
#include <iostream>
using namespace std;

// function ka prototype declare kiya gaya hai
void greet();
```

```
// main function jahan program execution start hota hai
int main() {

    // function ko call kiya ja raha hai
    greet();

    return 0;
}

// function ki definition jahan greeting message display hota hai
void greet() {

    // yahan message print kiya ja raha hai
    cout << "Welcome to Functions in C++ Lab" << endl;
}
```

Task 2: Function With Prototype and Call

```
#include <iostream>
using namespace std;

// function ka prototype declare kiya gaya hai
void sayHello();

// main function
int main() {

    // function call kiya ja raha hai
    sayHello();

    return 0;
}

// function definition
void sayHello() {

    // yahan Hello World message display ho raha hai
    cout << "Hello, World!" << endl;
}
```

Task 3: Function With Parameters and Return Value

```
#include <iostream>
using namespace std;

// function ka prototype declare kiya gaya hai
int add(int, int);
```

```
// main function
int main() {

    int result;

    // function ko arguments ke sath call kiya ja raha hai
    result = add(3, 5);

    // result display kiya ja raha hai
    cout << "Result: " << result << endl;

    return 0;
}

// function definition jo do numbers ko add karti hai
int add(int num1, int num2) {

    // dono numbers ka sum return kiya ja raha hai
    return num1 + num2;
}
```

5. Conclusion

In this lab, the concept of functions in C++ was implemented successfully. The tasks demonstrated function declaration, use of prototypes, function calls, and passing parameters with return values.